

Long chain n-3 fatty acids intake, fish consumption and suicide in a cohort of Japanese men and women--the Japan Public Health Center-based (JPHC) ...

OBJECTIVE: Eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) have been implicated as protective against suicide. However, it is uncertain whether a higher intake of EPA and DHA or of fish, a major source of these nutrients, lowers suicidal risk among Japanese, whose fish consumption and suicide rate are both high. This study prospectively examined the relation between fish, EPA, or DHA intake and suicide among Japanese men and women. **METHOD:** Subjects were 47,351 men and 54,156 women aged 40-69 years who participated in the JPHC Study, completed a food frequency questionnaire in 1995-1999, and were followed for death through December 2005. We used the Cox proportional hazards regression model to estimate the hazard ratio (HR) and 95% confidence interval (CI) for suicide by quintile of intake. **RESULTS:** A total of 213 and 85 deaths from suicide were recorded during 403,019 and 473,351 person-years of follow-up for men and women, respectively. Higher intakes of fish, EPA, or DHA were not associated with a lower risk of suicide. Multivariate HRs (95% CI) of suicide death for the highest versus lowest quintile of fish consumption were 0.95 (0.60-1.49) and 1.20 (0.58-2.47) for men and women, respectively. A significantly increased risk of suicidal death was observed among women with very low intake of fish, with HRs (95% CI) for those in 0-5th percentile versus middle quintile of 3.41 (1.36-8.51). **CONCLUSIONS:** Our overall result does not support a protective role of higher intake of fish, EPA, or DHA against suicide in Japanese men and women. Copyright © 2010 Elsevier B.V. All rights reserved.